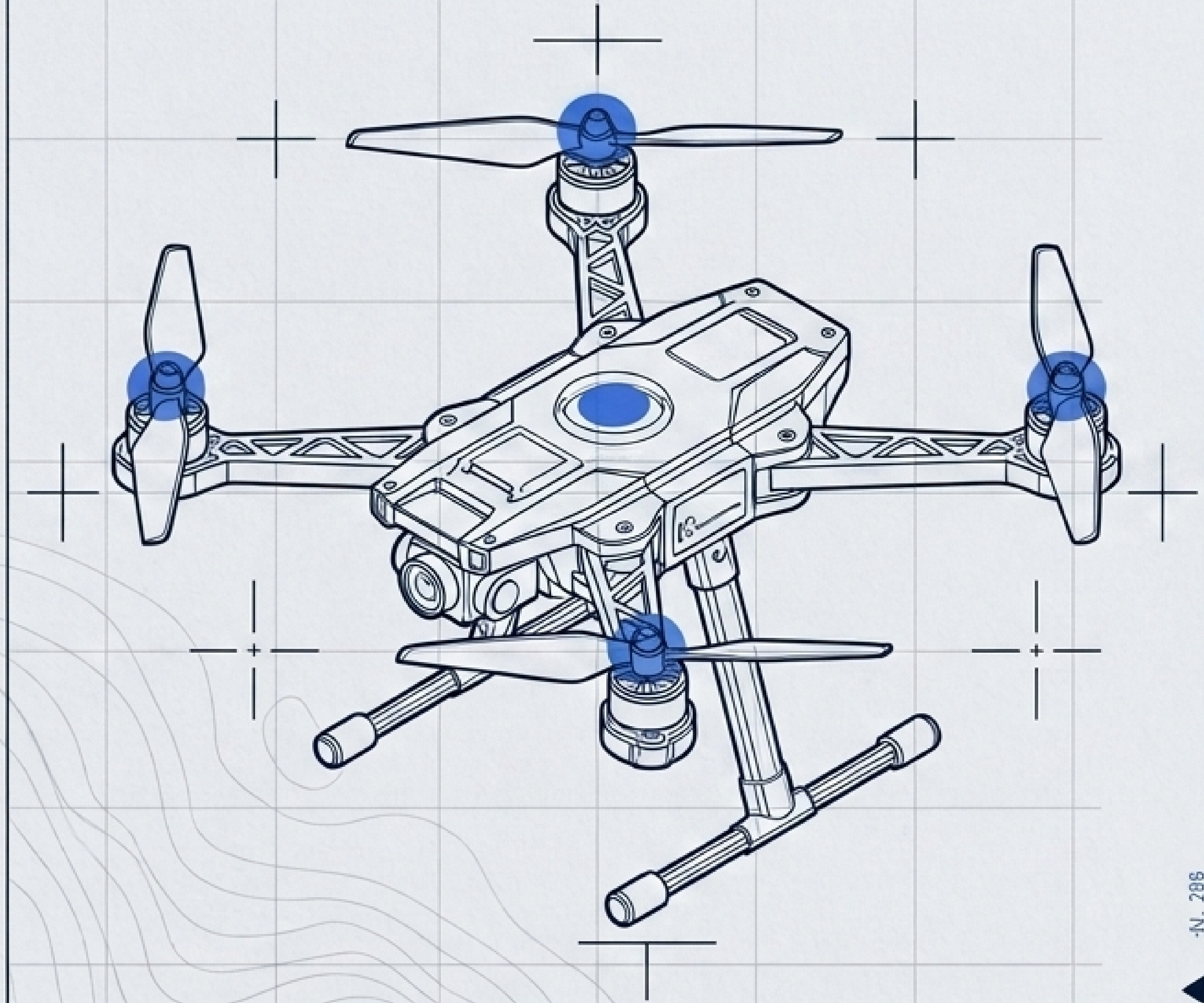


INTELLIGENT DRONE ROUTE PLANNING ALGORITHM

Batch: 2023-27

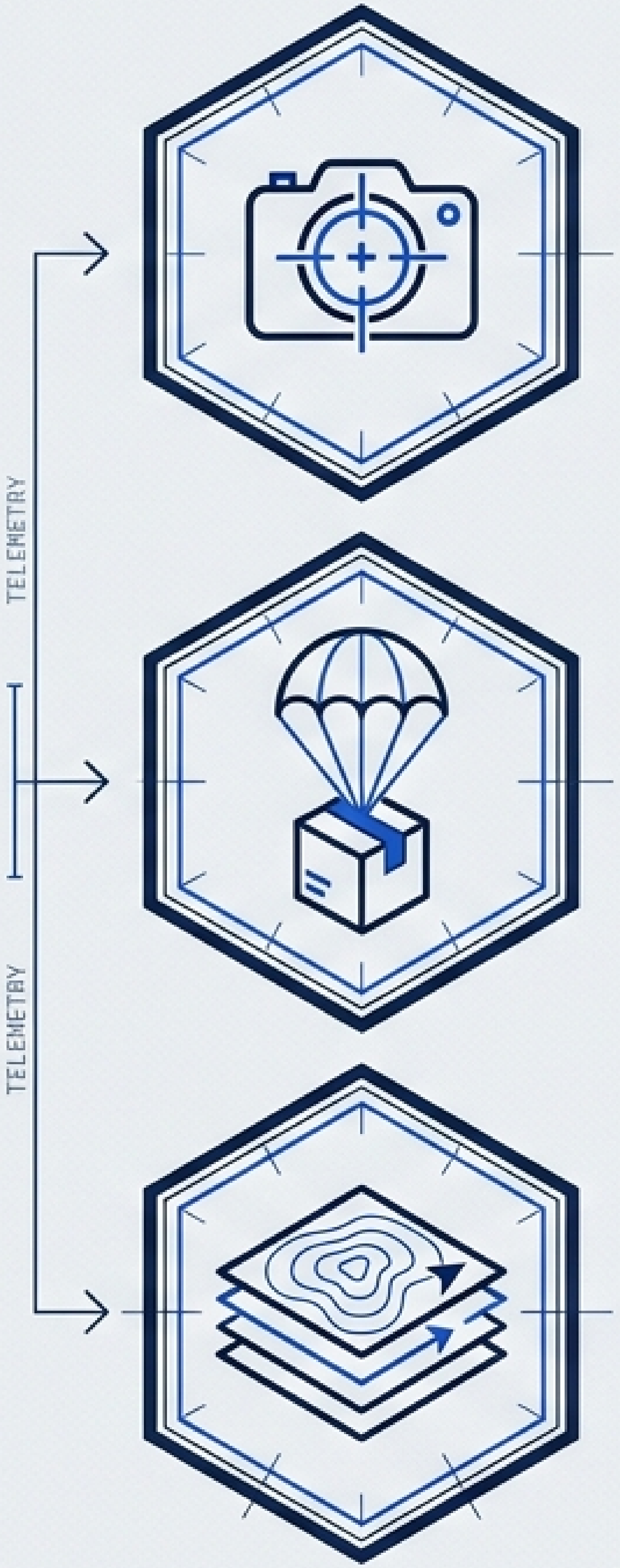
NAME	ROLL NO	SECTION
Siddharth Gaur	2301921540184	CSDS-4
Vaibhav Pachauri	2301921540206	CSDS-4
Sharad Pratap Singh	2301921540169	CSDS-4
Shivang Pandey	2301921540177	CSDS-4
Shaurya Pratap Singh	2301921540170	CSDS-4

The Landscape & The Stakes

UAV Route Planning

The computational process of determining a collision-free sequence of waypoints from origin to destination.

Static maps fail in dynamic worlds. True autonomy requires real-time pathfinding adaptation.

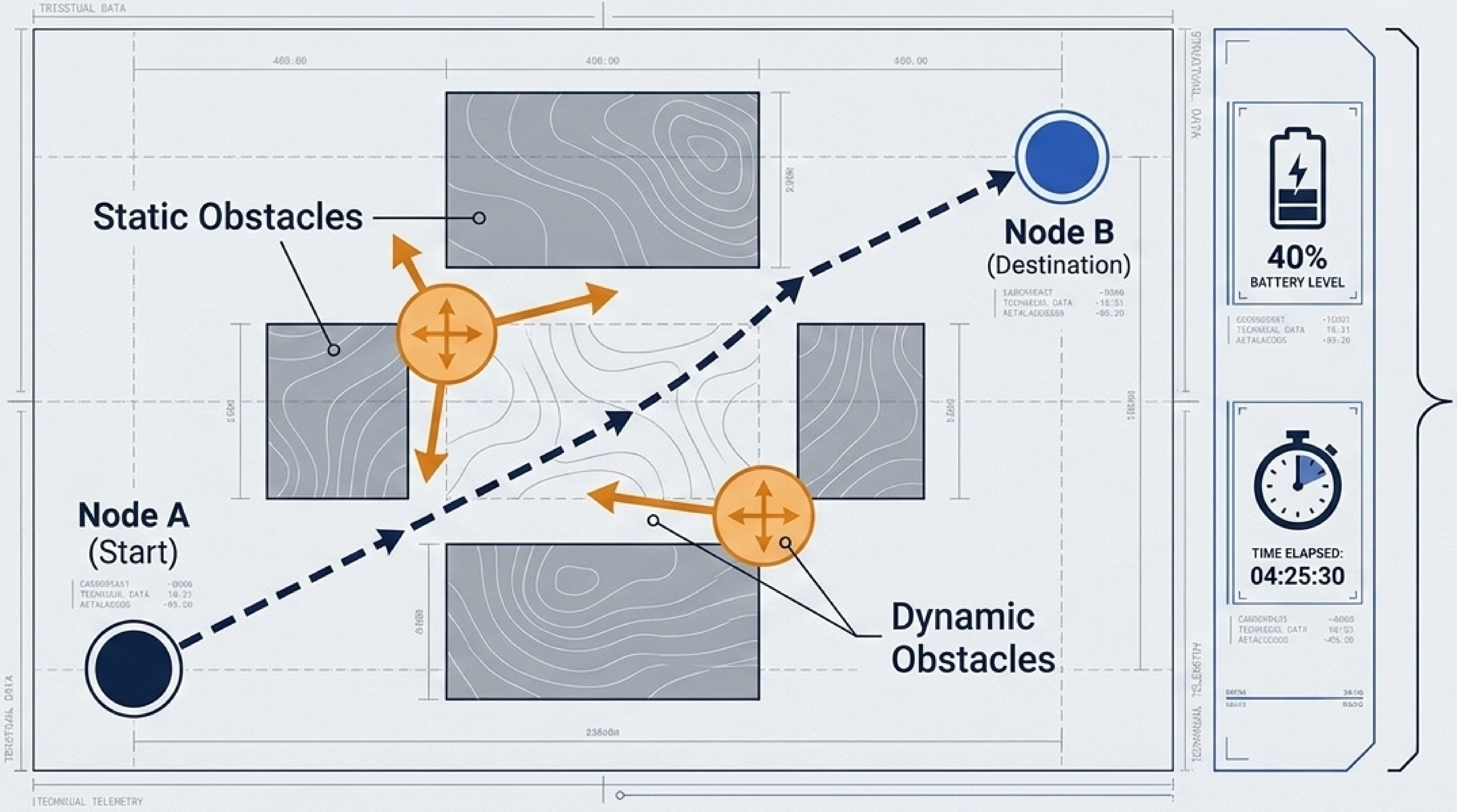


Surveillance

Package Delivery

Mapping

Problem Statement: Overlapping Constraints

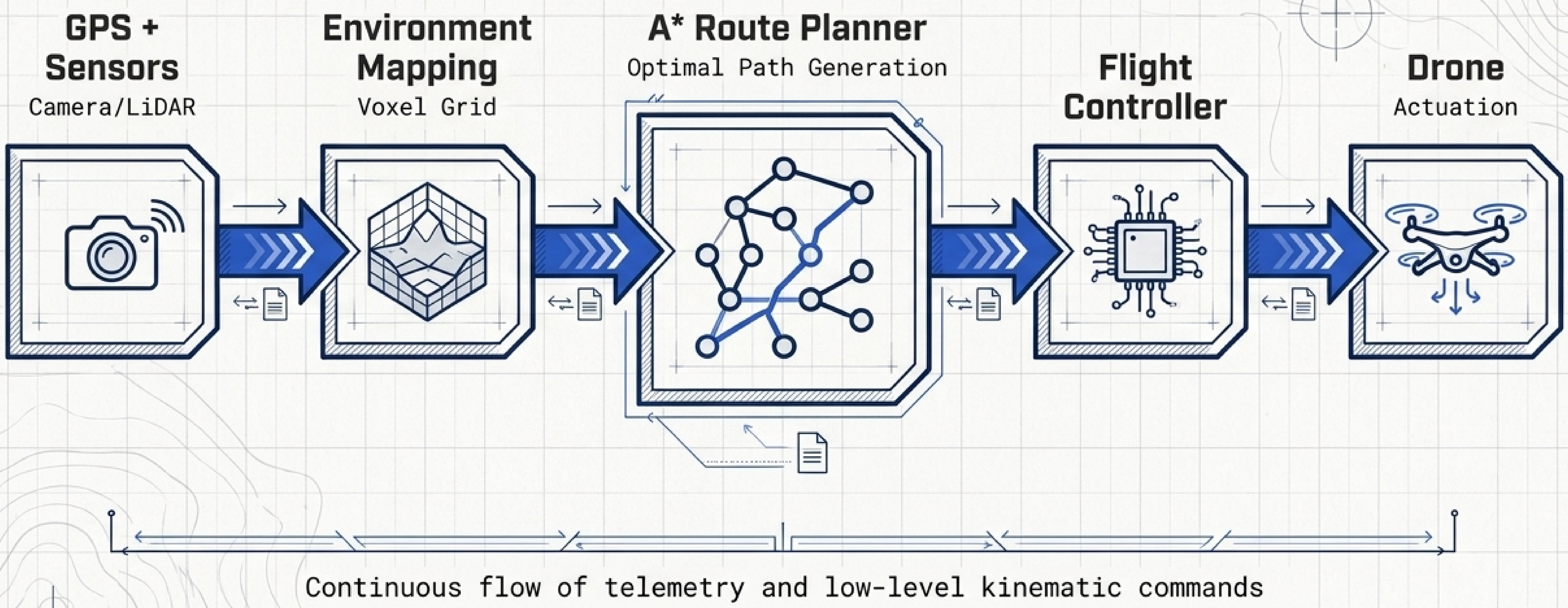


Require minimizing energy consumption and travel time while adapting dynamically.

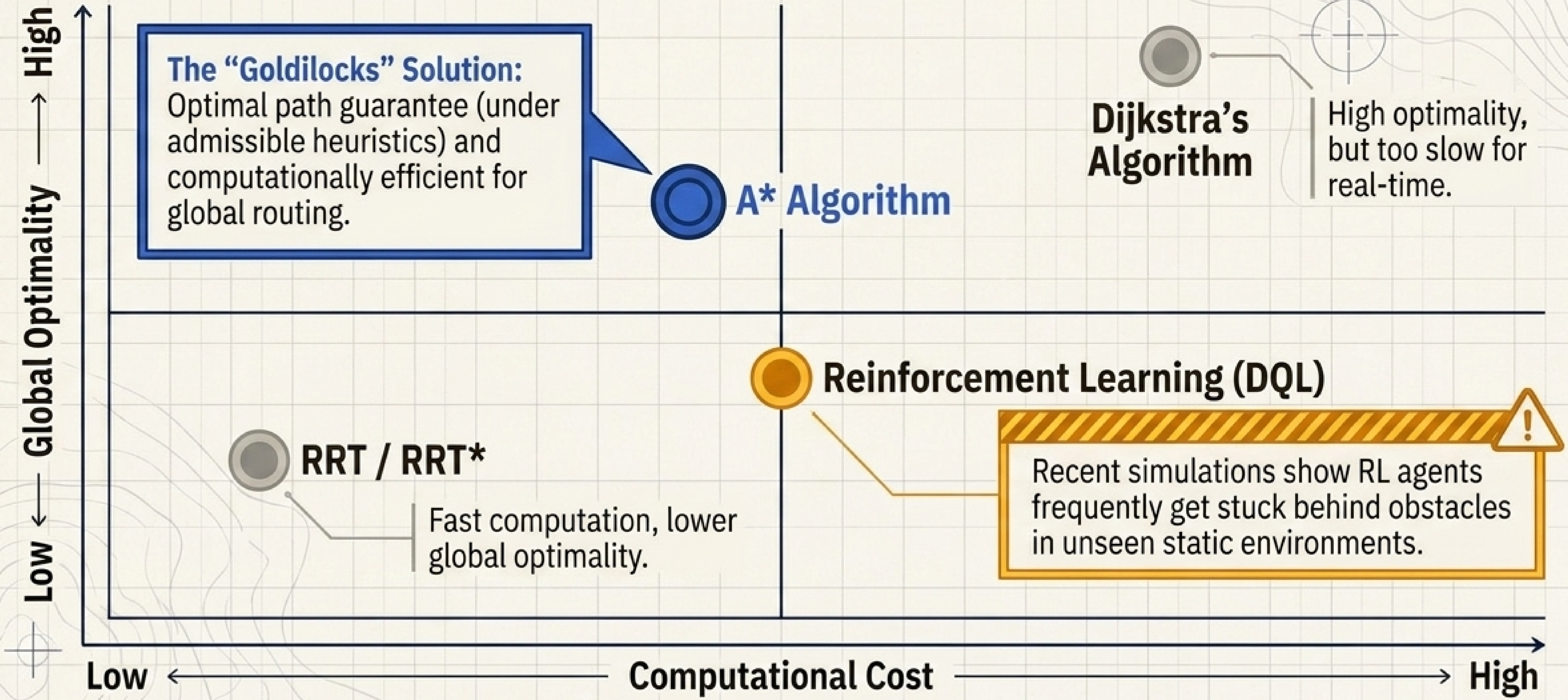
Paradigm Shift: Traditional vs Intelligent Route Planning

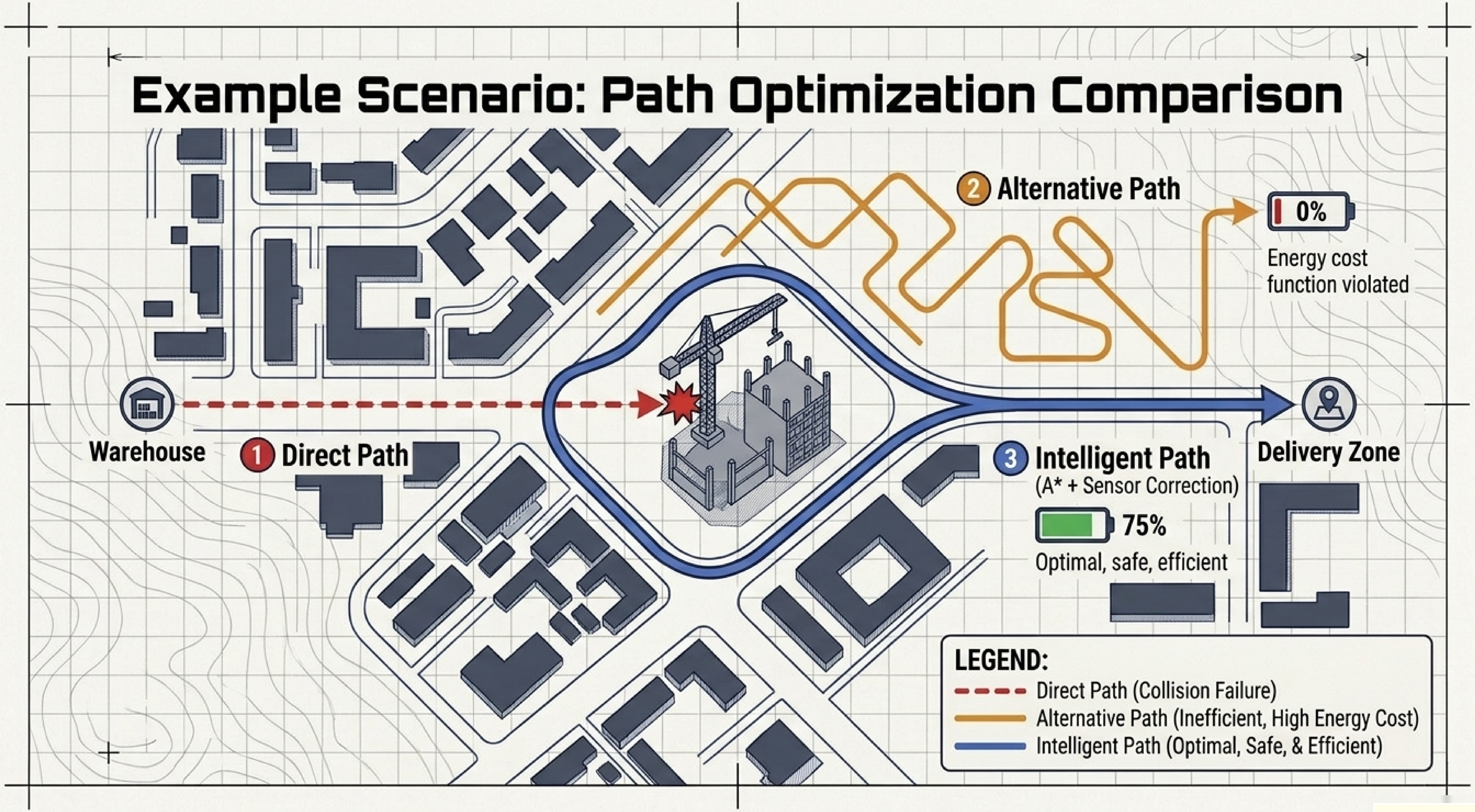
	Traditional Route Planning	Intelligent Route Planning
Route Adaptability	Fixed, pre-programmed routes	**Dynamic, on-the-fly trajectory generation** ✓
Obstacle Handling	Limited to known, static obstacles	**Real-time dynamic obstacle avoidance** ✓
Optimization Objectives	Distance-only optimization	**Multi-objective (Distance, Energy, Risk, Time)** ✓
Environmental Responsiveness	Lack of adaptiveness to shifting conditions	**Highly responsive to immediate sensor feedback** ✓

System Architecture & Information Flow



Route Planning Algorithms: The Landscape

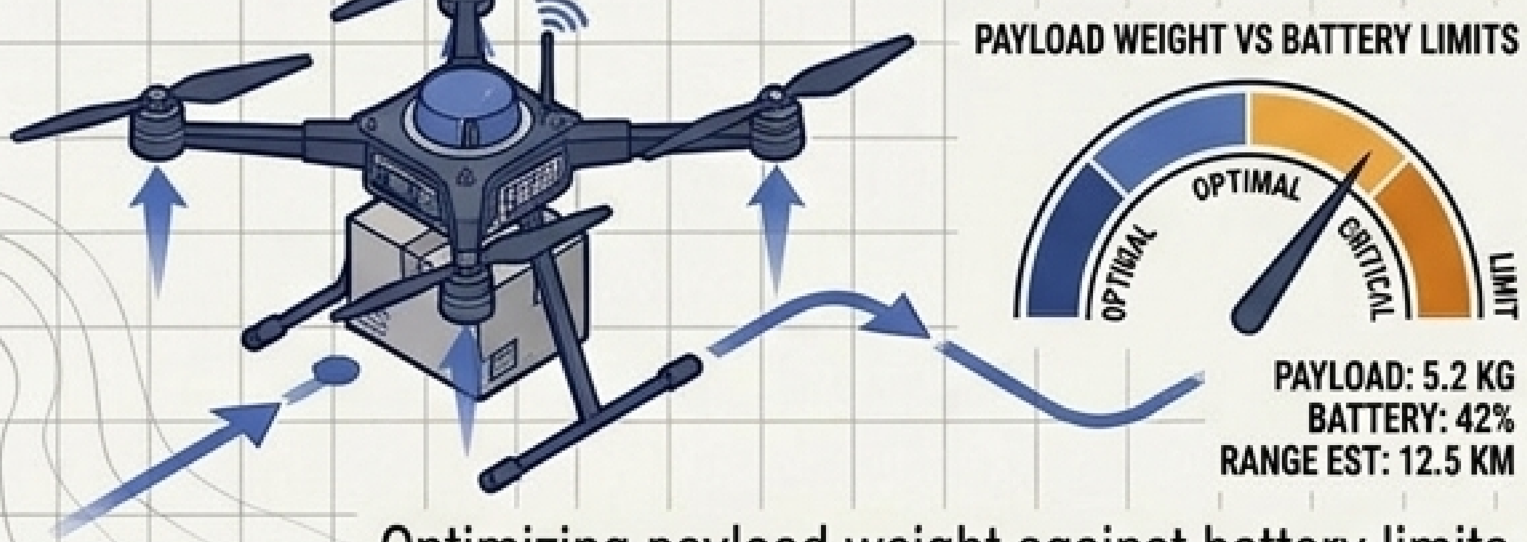




Example Scenario: Path Optimization Comparison
2 Alternative Path
0%
Energy cost
function violated
Warehouse
1 Direct Path
3 Intelligent Path
(A* + Sensor Correction)
75%
Optimal, safe, efficient
Delivery Zone
LEGEND:
Direct Path (Collision Failure)
Alternative Path (Inefficient, High Energy Cost)
Intelligent Path (Optimal, Safe, & Efficient)

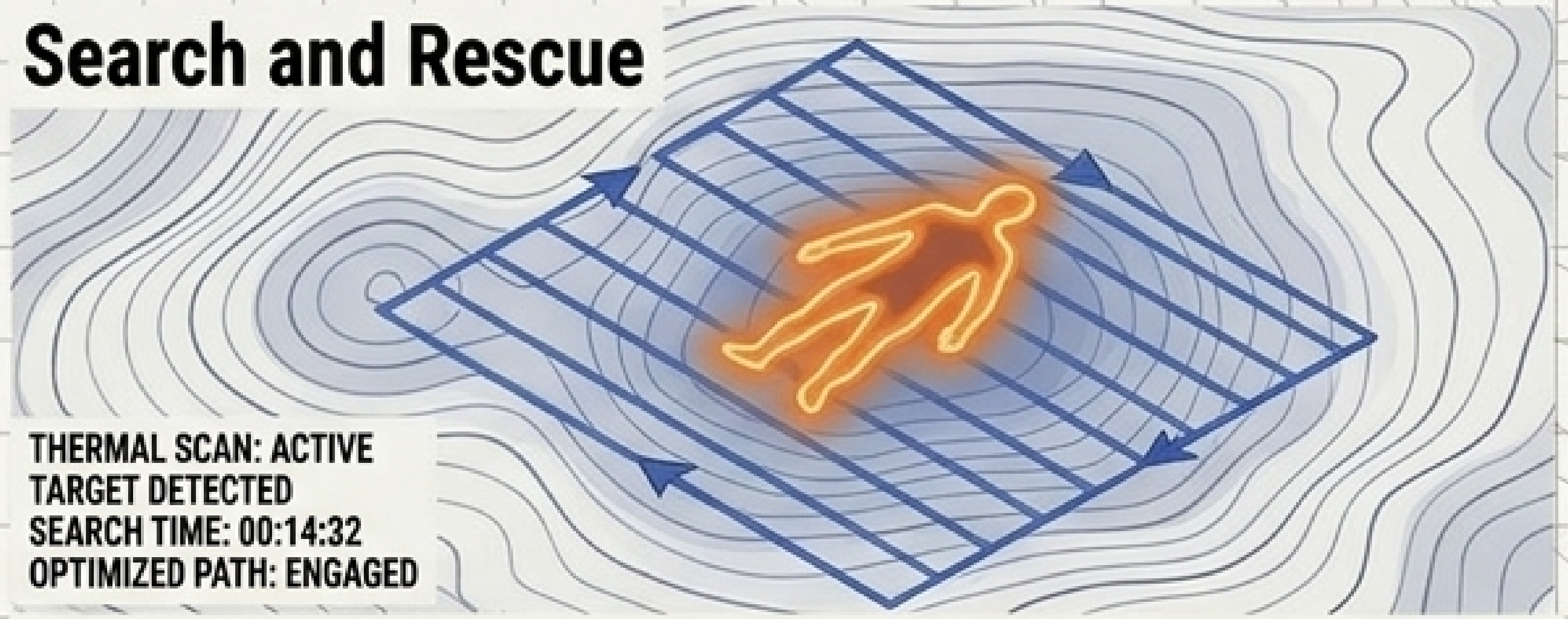
Diverse Applications & Real-World Utility

Package Delivery



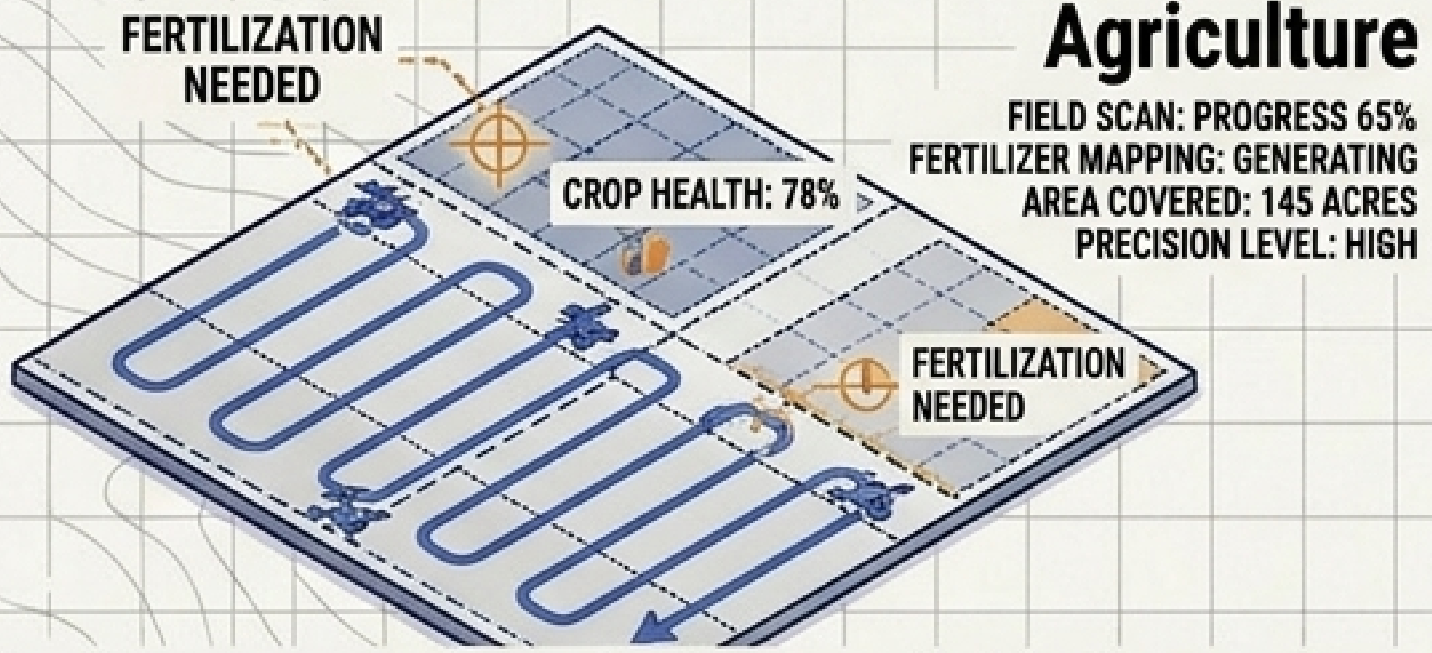
Optimizing payload weight against battery limits.

Search and Rescue



Utilizing thermal vision and minimizing search time in critical scenarios.

Agriculture



Precision crop scanning and autonomous fertilization mapping.

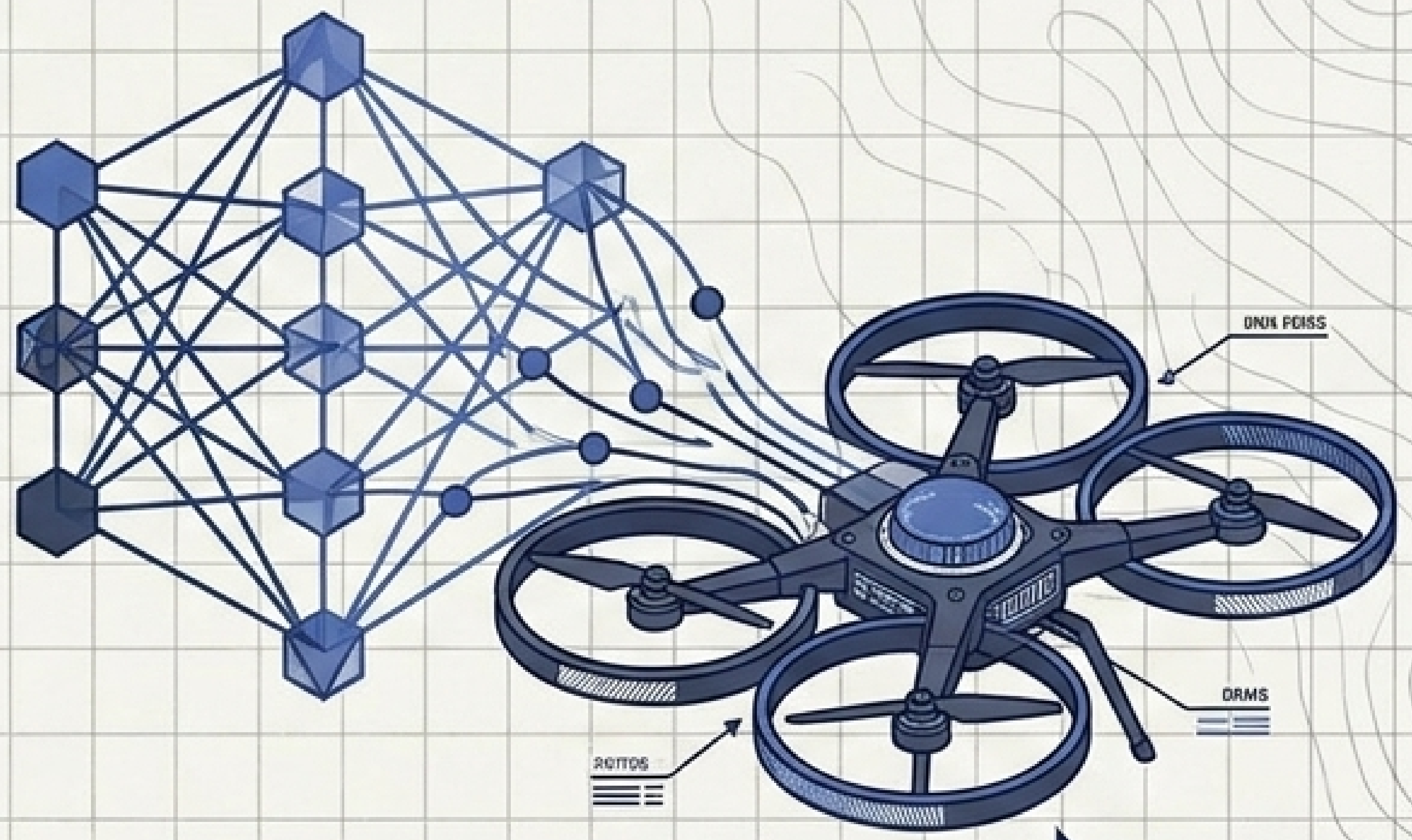
Disaster Management & Surveillance

Navigating compromised terrain with high risk-penalty routing.

Conclusion & The Next Frontier

UAV Autonomy Progress

- Demonstrated the foundational reliability and global optimality of the A* algorithm.
- Successfully integrated real-time sensor dynamic correction via hybrid frameworks to navigate unseen obstacles.
- Proved the flexibility of multi-objective cost functions for mission-specific adaptability.



Next Frontier: Integrating AI/ML for predictive environmental modeling and swarm intelligence.